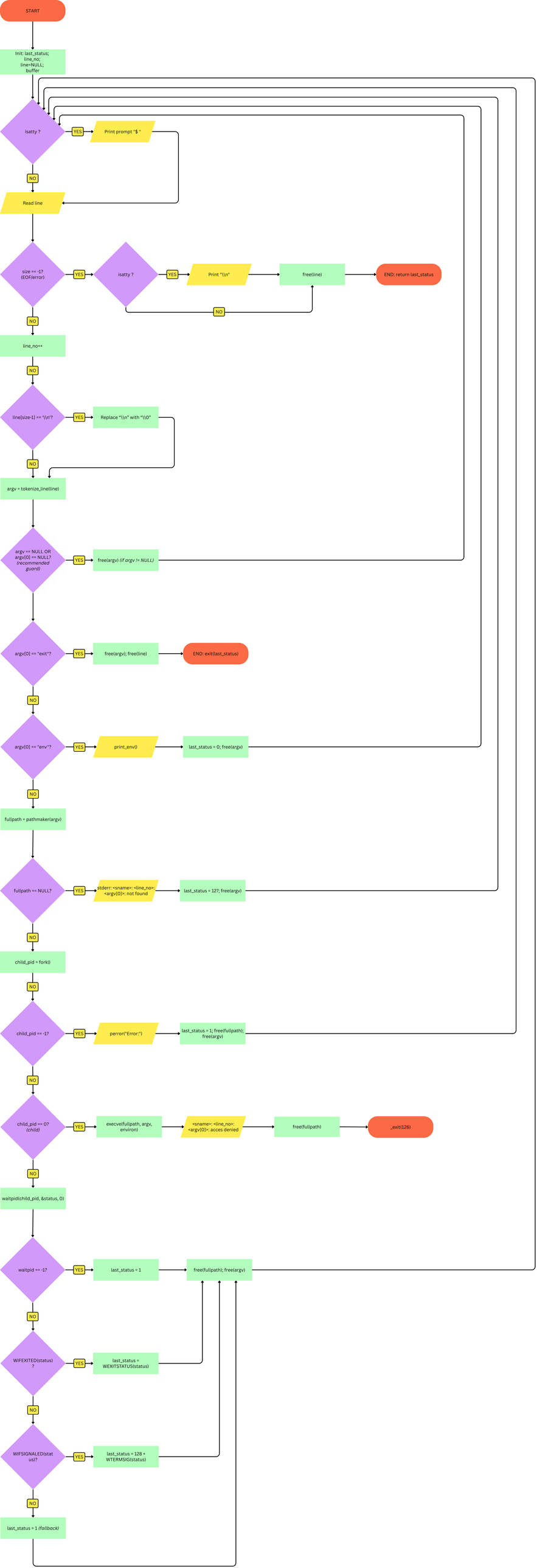
START
Init: last_status; line_no; line=NULL; buffer
isatty ?
YES
Print prompt "$ "
NO
Read line
size == -1? (EOF/error)
YES
isatty ?
YES
Print "\\n"
free(line)
END: return last_status
NO
NO
line_no++
NO
line[size-1] == "\\n"?
YES
Replace "\\n" with "\\0"
NO
argv = tokenize_line(line)
argv == NULL OR argv[0] == NULL? (recommended guard)
YES
free(argv) (if argv != NULL)
argv[0] == "exit"?
YES
free(argv); free(line)
END: exit(last_status)
NO
argv[0] == "env"?
YES
print_env()
last_status = 0; free(argv)
NO
fullpath = pathmaker(argv)
fullpath == NULL?
YES
stderr: <sname>: <line_no>: <argv[0]>: not found
last_status = 127; free(argv)
NO
child_pid = fork()
NO
child_pid == -1?
YES
perror("Error:")
last_status = 1; free(fullpath); free(argv)
NO
child_pid == 0? (child)
YES
execve(fullpath, argv, environ)
<sname>: <line_no>: <argv[0]>: acces denied
free(fullpath)
_exit(126)
NO
waitpid(child_pid, &status, 0)
waitpid == -1?
YES
last_status = 1
free(fullpath); free(argv)
NO
WIFEXITED(status) ?
YES
last_status = WEXITSTATUS(status)
NO
WIFSIGNALED(status)?
YES
last_status = 128 + WTERMSIG(status)
NO
last_status = 1 (fallback)